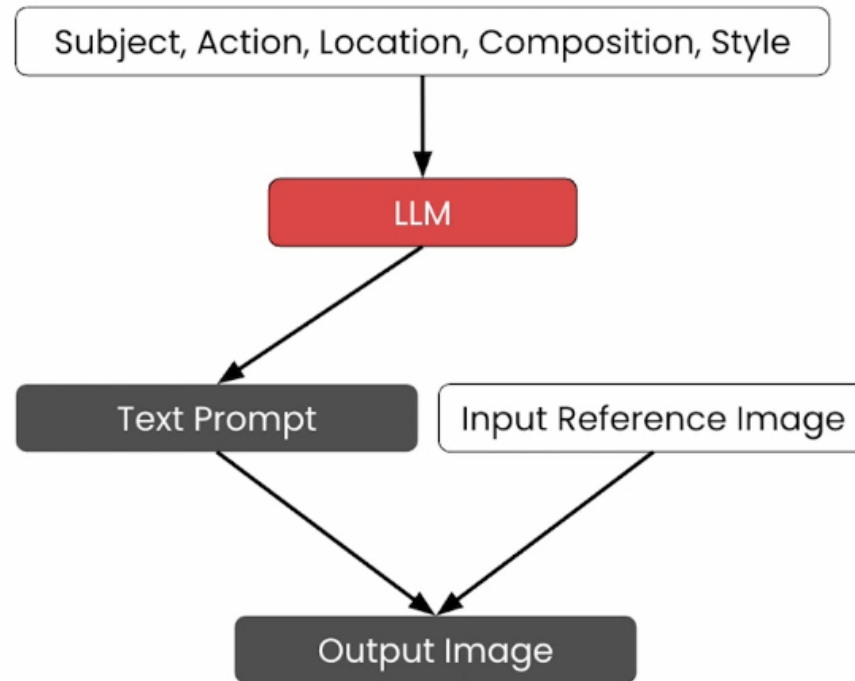


Image Prompt Attributes

- **Subject:** The “who” or “what” of the image
- **Action:** What’s happening
- **Location:** The “where” of the image
- **Composition:** Camera controls and lighting details
- **Style:** The overall aesthetic

Image Prompts



In the lab

1

Generate image from a basic text prompt

2

Formalize prompt structure

3

Reference stylized input images

4

Generate an image from an enhanced prompt

Let's see all of
this in the code

L2: Prompt Engineering for Image Generation

```
from IPython.display import Image, Markdown, display
from google import genai
from google.genai import types
```

```
import os
from helper import authenticate

credentials, project_id = authenticate()
client = genai.Client(
    project=project_id,
    location="global",
    credentials=credentials,
    http_options=types.HttpOptions(
        base_url=os.getenv("GOOGLE_VERTEX_BASE_URL")
    )
)
```

```
IMAGE_MODEL_ID = "gemini-3.1-flash-image-preview"
TEXT_MODEL_ID = "gemini-3-flash-preview"
```



Simple prompt

```
prompt = """
A diagram illustrating visual proof of the Pythagorean theorem.
"""

response = client.models.generate_content(
    model=IMAGE_MODEL_ID,
    contents=prompt,
    config=types.GenerateContentConfig(
        response_modalities=["IMAGE"],
        image_config=types.ImageConfig(
            # aspect ratios: 1:1, 3:2, 2:3, 3:4, 4:3, 1:4, 4:1, 4:5, 5:4, 1:8
            aspect_ratio="16:9",
        ),
    ),
)

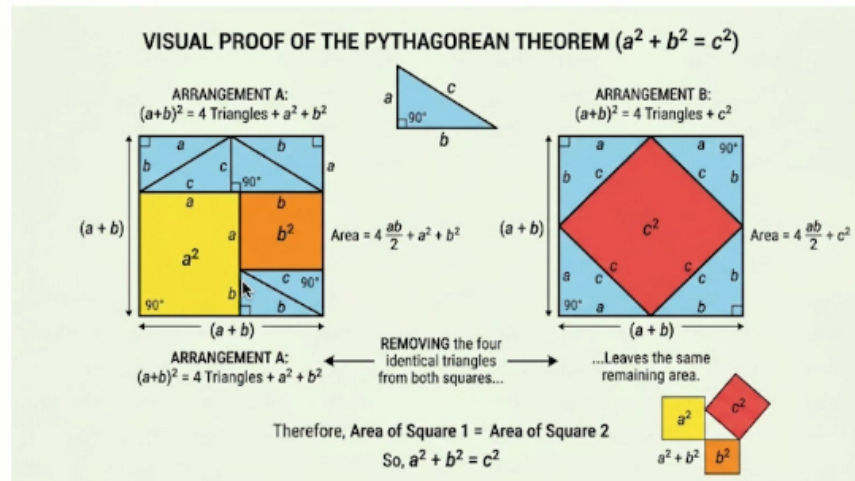
for part in response.candidates[0].content.parts:
    if part.inline_data:
        display(Image(data=part.inline_data.data, width=500))
```

```

response = client.models.generate_content(
    model=IMAGE_MODEL_ID,
    contents=prompt,
    config=types.GenerateContentConfig(
        response_modalities=["IMAGE"],
        image_config=types.ImageConfig(
            # aspect ratios: 1:1, 3:2, 2:3, 3:4, 4:3, 1:4, 4:1, 4:5, 5:4, 1:8
            aspect_ratio="16:9",
        ),
    ),
)

for part in response.candidates[0].content.parts:
    if part.inline_data:
        display(Image(data=part.inline_data.data, width=500))

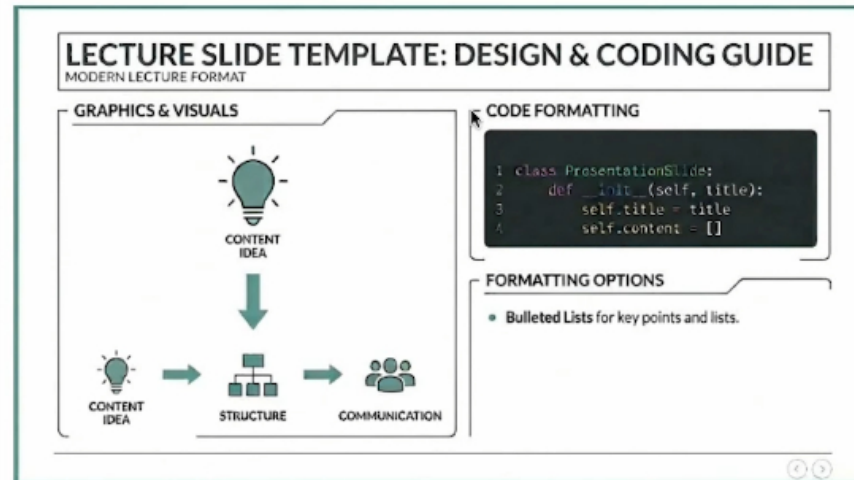
```



Prompt enhancement

```
subject = "a simple slide"
action = "explaining visual proof of the Pythagorean theorem"
location = "white background"
camera_control = "eye-level shot"
lighting = "white light"
style = "minimalist"
```

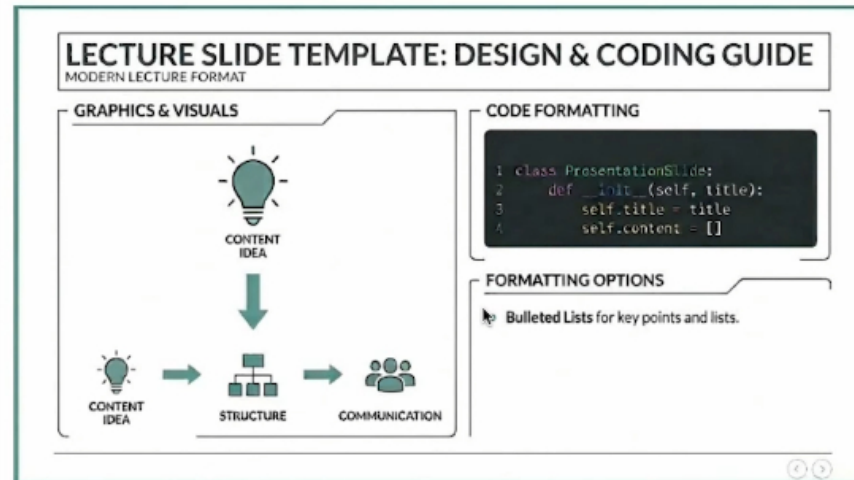
```
reference_image = "slide-template.png"
display(Image(filename=reference_image, width=500))
```



Prompt enhancement

```
subject = "a simple slide"
action = "explaining visual proof of the Pythagorean theorem"
location = "white background"
camera_control = "eye-level shot"
lighting = "white light"
style = "minimalist"
```

```
reference_image = "slide-template.png"
display(Image(filename=reference_image, width=500))
```



```
keywords = [subject, action, location, camera_control, lighting, style]

gemini_prompt = f"""
Your task is to expand the following keywords into a single, high-fidelity,
descriptive prompt for image generation. Every single keyword MUST be
included. Include reference images if provided and use that image as a
reference style guide for generated images. Output ONLY the final prompt
string, without any introduction or explanation. Mandatory Keywords:
{",".join(keywords)}
"""

response = client.models.generate_content(
    model=TEXT_MODEL_ID,
    contents=gemini_prompt,
)

image_prompt = response.text
display(Markdown(response.text))
```

A high-fidelity minimalist presentation slide providing a clear visual proof of the Pythagorean theorem, featuring a perfectly rendered right-angled triangle with squares built on each side to represent area calculations, captured in a clean eye-level shot against a stark white background, illuminated by soft neutral white light, emphasizing sharp geometric precision and simple educational design.

```
with open(reference_image, "rb") as f:
    image = f.read()

response = client.models.generate_content(
    model=IMAGE_MODEL_ID,
    contents=[
        types.Part.from_bytes(
            data=image,
            mime_type="image/png",
        ),
        f""""Use this reference image as a general style guide to generate a 1
slide. Include only the necessary sections from the template. Use the
as a general style formatting template instead of a strict outline. T
slide image should be the entire image. Slide concept: {image_prompt}
""",
    ],
    config=types.GenerateContentConfig(
        response_modalities=['IMAGE'],
        image_config=types.ImageConfig(
            aspect_ratio="16:9",
        ),
    ),
)

for part in response.candidates[0].content.parts:
    if part.inline_data:
        image_data=part.inline_data.data
        display(Image(data=image_data, width=500))

with open("slide-image.png", "wb") as image_file:
    image_file.write(image_data)
```

```

with open(reference_image, "rb") as f:
    image = f.read()

response = client.models.generate_content(
    model=IMAGE_MODEL_ID,
    contents=[
        types.Part.from_bytes(
            data=image,
            mime_type="image/png",
        ),
        f""""Use this reference image as a general style guide to generate a 1
slide. Include only the necessary sections from the template. Use the
as a general style formatting template instead of a strict outline. T
slide image should be the entire image. Slide concept: {image_prompt}
""",
    ],
    config=types.GenerateContentConfig(
        response_modalities=['IMAGE'],
        image_config=types.ImageConfig(
            aspect_ratio="16:9",
        ),
    ),
)

for part in response.candidates[0].content.parts:
    if part.inline_data:
        image_data=part.inline_data.data
        display(Image(data=image_data, width=500))

with open("slide-image.png", "wb") as image_file:
    image_file.write(image_data)

```

```
for part in response.candidates[0].content.parts:
    if part.inline_data:
        image_data=part.inline_data.data
        display(Image(data=image_data, width=500))

with open("slide-image.png", "wb") as image_file:
    image_file.write(image_data)
```

LECTURE SLIDE: GEOMETRIC PROOF

PYTHAGOREAN THEOREM DERIVATION

GEOMETRIC LAYOUT & PRINCIPLES

FUNDAMENTAL RELATIONSHIP:
 $a^2 + b^2 = c^2$

CONCEPT: AREAS → STRUCTURED AREA SUM → GEOMETRIC UNDERSTANDING

VISUAL DEMONSTRATION

KEY TAKEAWAYS

- $\text{Area}(a) + \text{Area}(b) = \text{Area}(c)$ for any right-angled triangle.
- The theorem states $a^2 + b^2 = c^2$.
- Confirmed by area rearrangement.

